Solution

We can use the equation

$$T_f = \frac{m_1 s_1 T_1 + m_2 s_2 T_2}{m_1 s_1 + m_2 s_2}$$

$m_1 = 5L = 5\text{kg}$ and $m_2 = 4L = 4\text{kg}$, $s_1 = s_2$ and $T_1 = 50°\text{C} = 323\text{K}$ and $T_2 = 30°\text{C} = 303\text{ K}$.

So

$$T_f = \frac{m_1 T_1 + m_2 T_2}{m_1 + m_2} = \frac{5 \times 323 + 4 \times 303}{5 + 4} = 314.11\text{ K}$$

$T_f = 314.11\text{ K} - 273\text{K} \approx 41°\text{C}$.

Suppose if we mix equal amount of water ($m_1 = m_2$) with 50°C and 30°C, then the final temperature is average of two temperatures.

$$T_f = \frac{T_1 + T_2}{2} = \frac{323 + 303}{2} = 313\text{K} = 40°\text{C}$$

Suppose if both the water are at 30°C then the final temperature will also 30°C. It implies that they are at equilibrium and no heat exchange takes place between each other.

Note

It is important to note that the final equilibrium temperature of mixing of gas or liquid depends on mass of the substances, their specific heat capacities and their temperatures. Only if we mix the same substances at equal amount, the final temperature will be an average of the individual temperatures.

8.2.7 Heat transfer

As we have seen already heat is a energy in transit which is transferred from one body to another body due to temperature difference. There are three modes of heat transfer: Conduction, Convection and Radiation.

Conduction

Conduction is the process of direct transfer of heat through matter due to temperature difference. When two objects are in direct contact with one another, heat will be transferred from the hotter object to the colder one. The objects which allow heat to travel easily through them are called conductors.

Thermal conductivity

Thermal conductivity is the ability to conduct heat.

The quantity of heat transferred through a unit length of a material in a direction normal to unit surface area due to a unit temperature difference under steady state conditions is known as thermal conductivity of a material.

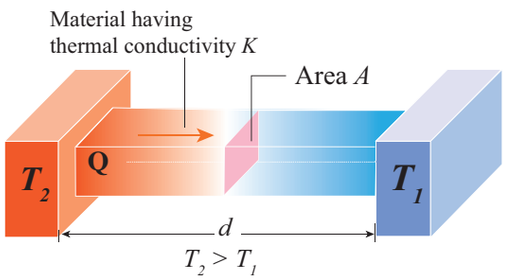


Figure 8.11 Steady state heat flow by conduction.

In steady state, the rate of flow of heat Q is proportional to the temperature difference ΔT and the area of cross section A and is inversely proportional to the length L. So the rate of flow of heat is written as

$$\frac{Q}{t} = \frac{KA\Delta T}{L}$$

Where, K is known as the **coefficient of thermal conductivity.**
(Not to be confused with Kelvin represented by upper case K)

The SI unit of thermal conductivity is

$$\text{J s}^{-1}\text{ m}^{-1}\text{ K}^{-1} \text{ or W m}^{-1}\text{ K}^{-1}.$$

UNIT-8(XI-Physics_Vol-2).indd 107 02-01-2023 15:25:32